FAULTGAT: A Graph Attention Network-Based Approach for Intermittent Fault Diagnosis in Multiprocessor Systems

Xueli Sun, Shuangxiang Kan, Jianxi Fan

***Abstract*—In the realm of interconnection networks, reliability analysis is of utmost importance, especially as component vulnerability increases with system scale. Among various fault types, intermittent faults—sporadic and transient in nature—are particularly challenging to detect and diagnose. Traditional diagnostic algorithms work well only within small theoretical diagnosability bounds and with complete test data, making them ineffective in practical scenarios involving incomplete syndromes or high fault rates. Recent neural network-based methods offer improvements but often ignore the local nature of faults and demand large training datasets, limiting their scalability and real-world applicability. In this work, we propose FAULTGAT, a novel Graph Attention Network (GAT)-based framework designed for intermittent fault diagnosis under the PMC model. By leveraging the locality-aware structure of GATs, FAULTGAT performs node-level diagnosis using only local syndrome information, aligning closely with the diagnostic principles of the PMC model. This approach improves scalability, enhances robustness to missing data, and reduces dependence on large training datasets. Experimental evaluations across diverse network topologies demonstrate that FAULTGAT outperforms existing methods, achieving nearly 100% accuracy within theoretical diagnosability bounds while maintaining over 90% accuracy even when the number of faulty nodes exceeds theoretical limits or when operating with partial syndromes.**

***Index Terms*—Fault tolerance; PMC model; intermittent fault diagnosis; partial syndromes; graph attention networks**

I. INTRODUCTION

Fault diagnosis plays a critical role in ensuring the reliability and availability of complex computing systems and interconnection networks. As systems grow in scale and complexity, the ability to accurately identify malfunctioning components becomes essential for maintaining operational integrity and preventing cascading failures. Effective diagnostic mechanisms allow for timely intervention, reducing downtime and maintenance costs while preserving overall system performance [1]–[3]. Among the various fault types that can affect these systems, intermittent faults present particularly challenging diagnostic scenarios. Unlike permanent faults that persist consistently, intermittent faults are characterized by their sporadic and transient nature, appearing and disappearing during the testing process. This temporal variability creates complex situations where test results may change over time for the same processor pair, making traditional diagnostic approaches less effective. The intermittent fault diagnosability of a system, defined as the maximum number of faulty processors that can be correctly identified under these conditions, serves as a crucial metric for evaluating system reliability in the presence of such elusive faults.

Traditional diagnosis algorithms can effectively identify faulty nodes when the number of faulty processors remains within the theoretical diagnosability bounds, which are typically quite small [4]–[7]. In addition, these algorithms face significant limitations in practical scenarios. When the number of faulty processors exceeds the diagnosability threshold, traditional algorithms cannot guarantee correct identification of all faulty nodes, often resulting in ambiguous or incorrect diagnoses. Furthermore, these conventional approaches typically require complete syndrome information to function properly [8]–[11]. In real-world environments, where test results may be partially missing or corrupted due to communication failures or testing limitations, traditional algorithms often fail to provide reliable diagnoses. This gap between theoretical guarantees and practical requirements has motivated researchers to explore alternative approaches.

In recent years, deep learning approaches have been explored for system-level fault diagnosis to overcome these limitations. Elhadef [12] pioneered an ANN-based algorithm under the PMC model for permanent fault diagnosis, accelerating diagnosis while maintaining efficacy in large-scale systems. Subsequently, Mourad and Nayak [13] extended this paradigm to comparison-based models, demonstrating that multilayer feedforward neural networks can effectively identify faulty nodes even with partial syndromes. Their work showed that neural networks can learn to recognize patterns in test outcomes that indicate the presence of faults, providing a viable alternative to traditional diagnosis algorithms. Building on this foundation, Lin et al. [14] proposed a similar neural network enabled approach for intermittent fault diagnosis under the comparison model. Their work also utilized a multilayer feedforward neural network to process partial syndrome information. They established the intermittent fault diagnosability of regular networks under the comparison model and developed an algorithm to solve the fault identification problem with incomplete test information. This work represents an advancement in addressing the challenges of intermittent fault diagnosis with incomplete test information, particularly in the comparison model context.

While previous neural network approaches have shown

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promising results, they suffer from significant limitations. Both [13] and [14] approach train their models using the entire graph's syndrome data as input, failing to effectively leverage the localized nature of fault information in the comparison model. In the comparison model, a node's status (faulty or fault-free) primarily depends on test results involving only that node and its immediate neighbors. By processing the entire network's syndrome data collectively, these methods may introduce irrelevant information as noise, potentially degrading diagnostic accuracy. Furthermore, these approaches require extensive training datasets to achieve reasonable performance, limiting their practical applicability in dynamic network environments.

In this paper, we propose FAULTGAT, a novel framework that leverages Graph Attention Networks (GATs) to determine the fault status of each node under the PMC model for intermittent fault diagnosis. Unlike previous neural network approaches that process the entire graph's diagnosis syndrome and introduce irrelevant information as noise, GATs focus on each node's local neighborhood information and use attention mechanisms to dynamically weight the importance of different neighboring nodes and their associated test results [15]. This locality-aware processing with selective attention aligns with the PMC model's characteristics [16], where test outcomes primarily reflect local fault conditions rather than global network states. To address the fundamental challenge that intermittent faulty nodes may appear normal during single-round testing (making them indistinguishable from truly fault-free nodes), FAULTGAT employs multi-round testing and designs proportion-based statistical node features that capture the sporadic nature of intermittent faults. By aggregating test results across multiple rounds into statistical patterns, FAULTGAT transforms the temporal uncertainty of intermittent faults into detectable features for GAT-based classification. This approach maintains the locality principle inherent in the PMC model while effectively distinguishing between fault-free processors and those experiencing intermittent faults.

We conducted extensive experiments to validate the performance of FAULTGAT across various network topologies and fault scenarios. Specifically, we evaluated FAULTGAT on two representative interconnection network topologies: the hypercube Q_n and the augmented k-ary n-cube $AQ_{n,k}$, which encompass both simple and complex graph structures. We demonstrated that FAULTGAT consistently outperforms existing neural network approaches. Within theoretical diagnosability bounds, FAULTGAT achieves near 100% accuracy and F1-scores, correctly identifying all faulty nodes with high precision. Beyond theoretical diagnosability bounds, FAULTGAT exhibits great robustness, maintaining accuracy above 90% even when the proportion of intermittent faulty nodes reaches 50% of the network—a scenario where traditional algorithms fail. Notably, FAULTGAT also demonstrates exceptional resilience to partial or incomplete syndromes, maintaining over 90% accuracy even with 50% of nodes disabled. These results highlight FAULTGAT's practical applicability in real-world intermittent fault scenarios.

In summary, our main contributions are as follows:

- We propose FAULTGAT, a novel Graph Attention Network-based framework for intermittent fault diagnosis that leverages the locality-aware structure of GATs to perform node-level diagnosis using only local syndrome information, aligning with the diagnostic principles of the PMC model.
- We introduce a multi-round testing and statistical feature extraction approach that effectively transforms the sporadic nature of intermittent faults into learnable patterns, enabling accurate diagnosis even when fault counts exceed theoretical diagnosability bounds.
- We demonstrate through extensive experiments that FAULTGAT significantly outperforms existing methods across various network topologies, achieving superior accuracy with less training data and maintaining reliable performance under challenging conditions including high fault ratios and incomplete syndromes.

The artifact for our work is publicly available at: https://github.com/shuangxiangkan/FaultGAT.git.

II. PRELIMINARIES

In this section, we introduce the key concepts and terminology related to the PMC diagnostic model, intermittent fault diagnosis, and the application of graph attention networks for fault node classification in multiprocessor systems. The unspecified terminology follows the conventions in [17].

A. The PMC Diagnostic Model

The *PMC model* [16] is a system-level diagnostic framework where processors test each other to identify faults. The PMC model can be considered a special case of the comparison model, where each processor directly tests its neighbors rather than comparing the responses of two neighbors to the same input. In this model, a multiprocessor system is represented by a directed testing digraph $\mathcal{D} = (V, E)$, where V is the set of processors (also referred to as nodes), and a directed edge $(u, v) \in E$ indicates that processor u tests processor v.

The PMC model relies on mutual testing, where each test generates a binary outcome based on the fault status of the testing and tested processors. Only fault-free processors produce reliable test results, while faulty processors may yield arbitrary outcomes. The collection of these test outcomes, known as the *syndrome*, forms the basis for diagnosing faulty processors.

Definition 1. In the PMC model, for permanent faults, a *syndrome* is the collection of all test outcomes in the testing digraph $\mathcal{D} = (V, E)$, represented by a function $\sigma : E \to \{0, 1\}$, where $\sigma(u, v)$ denotes the binary result of processor u testing processor v. The syndrome serves as the input for identifying faulty processors in the system. The test outcomes are determined as follows:

- If both u and v are fault-free, then $\sigma(u, v) = 0$.
- If u is fault-free and v is faulty, then $\sigma(u, v) = 1$.
- If u is faulty, then $\sigma(u, v) \in \{0, 1\}$, regardless of the fault status of v.

B. Intermittent Faults

Intermittent faults, unlike permanent faults, appear or disappear during testing, leading to inconsistent test outcomes for the same processor pair. This variability complicates system-level fault diagnosis, as processors may produce differing results across testing rounds, generating inconsistent syndromes that challenge traditional permanent fault diagnosis algorithms. An intermittently faulty processor may temporarily behave as fault-free, becoming indistinguishable from fault-free processors in certain tests, while unreliable results from such processors add noise to the syndrome, obscuring fault patterns. Building upon the PMC model, Mallela et al. [18] introduced the following rules for intermittent fault diagnosis:

Definition 2. In the PMC model, for intermittent faults, a *syndrome* is the set of all test outcomes in the directed testing digraph $\mathcal{D} = (V, E)$, represented by a function $\sigma : E \to \{0, 1\}$, where $\sigma(u, v)$ denotes the binary result of processor u testing processor v. For a fault set $IF \subseteq V$ of intermittently faulty processors, the test outcomes are governed by the following rules:

- If both u and v are fault-free ($u \notin IF, v \notin IF$), then $\sigma(u, v) = 0$.
- If u is fault-free and v is intermittently faulty but temporarily fault-free ($u \notin IF, v \in IF$), then $\sigma(u, v) = 0$.
- If u is fault-free and v is intermittently faulty and active ($u \notin IF, v \in IF$), then $\sigma(u, v) = 1$.
- If u is intermittently faulty ($u \in IF$), then $\sigma(u, v) \in \{0, 1\}$, as the result is unreliable.

To quantify the diagnostic capability under intermittent faults, Mallela et al. [18] introduced the intermittent fault diagnosability t_i, which represents the maximum number of intermittently faulty processors a system can reliably handle while maintaining diagnostic accuracy.

Definition 3. A system S is t_i-fault diagnosable if, given no more than t_i intermittently faulty processors, the system can guarantee that no fault-free processor is misdiagnosed as faulty, and the diagnosis is at worst incomplete but never incorrect.

The introduction of intermittent faults elevates the computational complexity of fault diagnosis significantly beyond that of permanent faults. While permanent fault diagnosis is already NP-hard for general comparison-based models [19], intermittent fault diagnosis presents additional layers of complexity. When the number of intermittently faulty processors is bounded by t_i, polynomial-time algorithms can provide reliable diagnosis [18]. However, when this bound is exceeded, or when syndromes are incomplete due to testing failures or communication disruptions, traditional deterministic approaches become inadequate. Real-world scenarios often involve incomplete syndromes where not all test outcomes are available due to processors failing during testing or communication link failures. These missing test outcomes introduce additional ambiguity, preventing complete reconstruction of the fault set and further exacerbating the computational challenges.

C. Graph Attention Networks

To address these challenges, we propose the use of Graph Attention Networks (GATs) for intermittent fault diagnosis. GATs are neural networks designed for graph-structured data that use attention mechanisms to assign different weights to neighboring nodes, enabling the model to prioritize more relevant connections when aggregating information.

By modeling the fault diagnosis task as a binary node classification problem (faulty vs. fault-free), GATs leverage the graph structure of the testing digraph and employ attention mechanisms to dynamically weigh the importance of test outcomes from neighboring nodes based on their relevance. This approach captures complex dependencies in the system topology and exploits structural features, such as node degrees and connectivity patterns, enhancing the accuracy and robustness of fault node identification.

GATs are suitable for handling noisy or incomplete syndromes, as they can prioritize reliable connections and effectively distinguish faulty nodes from fault-free ones, even in the presence of inconsistent data. Furthermore, the offline learning capability of GATs reduces diagnosis latency, making them a promising solution for tackling the NP-hard intermittent fault diagnosis problem, especially in scenarios where traditional algorithms are infeasible due to incomplete syndromes or excessive faults.

D. Evaluated topologies

The FAULTGAT proposed in this paper using GATs for intermittent fault diagnosis under the PMC model is designed for general system structures. For evaluation, we focus on two representative topologies: the hypercube, a simple and well-studied structure, and the augmented k-ary n-cube, a more complex topology with enhanced connectivity. These choices are not cherry-picked but are selected to demonstrate the effectiveness of our approach across a range of system complexities, with comparable performance expected for other topologies. Below, we present these topologies and their intermittent fault diagnosability under the PMC model in advance.

Definition 4 (*Hypercube* [20])**.** An n-dimensional hypercube, also known as an n-cube and denoted by Q_n, is defined as follows:

$$Q_n = \{(x_1, x_2, \ldots, x_n) \mid x_i \in \{0, 1\} \text{ for } i = 1, 2, \ldots, n\}.$$

The hypercube Q_n has 2^n nodes. Two nodes are connected by an edge if and only if their Hamming distance is 1, i.e., they differ in exactly one coordinate (see Fig. 1, left).

Definition 5 (*Augmented k-ary n-cube* [21])**.** An Augmented k-ary n-cube, denoted by $AQ_{n,k}$, is a k-ary n-cube enhanced with additional edges to increase connectivity. The node set of $AQ_{n,k}$ consists of k^n nodes, each represented as an n-tuple $(x_1, x_2, \ldots, x_n)$ where $x_i \in \{0, 1, \ldots, k-1\}$ for $i = 1, 2, \ldots, n$. For $k \geq 3$, $AQ_{1,k}$ consists of nodes $\{0, 1, \ldots, k-1\}$, where nodes u and v are adjacent if and only if $v = u-1$ or $v = u+1$. When $n \geq 2$, $AQ_{n,k}$ is constructed from k copies of $AQ_{n-1,k}$, denoted as $AQ_{n,k} = AQ^0_{n,k} \cup AQ^1_{n,k} \cup \cdots \cup AQ^{k-1}_{n,k}$,

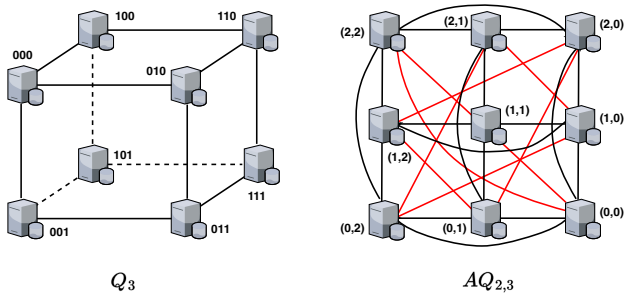

Fig. 1: The 3-dimensional hypercube Q_3 and Augmented 3-ary 2-cube $AQ_{2,3}$

where $AQ^i_{n,k} \cong AQ_{n-1,k}$ for $i \in [0, k-1]$. The ith copy is labeled by appending i as the nth coordinate to each node, ensuring all nodes within the same copy share an identical nth coordinate (see Fig. 1, right).

Under the PMC model, the intermittent fault diagnosability of the hypercube is $t_i(Q_n) = n - 2$ for $n \geq 3$, and that of the augmented k-ary n-cube is $t_i(AQ_{n,k}) = 4n - 6$ for $n \geq 4$ and $k \geq 4$, as can be derived from [21]–[23].

III. Methodology

This section presents our FaultGAT framework, a novel approach for intermittent fault diagnosis in multiprocessor systems under the PMC model. FaultGAT leverages Graph Attention Networks to address the challenges of detecting intermittent faults, which are characterized by their sporadic and transient nature. Figure 2 provides a high-level overview of the FaultGAT architecture, illustrating its three main phases: multi-round testing, locality-aware feature extraction, and GAT-based classification. The key insight is to transform the probabilistic nature of intermittent faults into statistical patterns that can be learned by GATs through locality-aware feature extraction.

A. Multi-Round Testing

The key challenge in intermittent fault diagnosis lies in the sporadic nature of fault manifestation. Unlike permanent faults that consistently exhibit faulty behavior, intermittent faults alternate between normal and faulty states unpredictably. This creates a fundamental problem for single-round testing approaches. Consider an intermittent faulty processor u_i (where subscript i denotes intermittent faulty node) that is currently manifesting fault-free status during a single test round. If all its neighbors are fault-free, the processor's feature vector will be:

$$\mathbf{f}^{single}_{u_i} = [0, 0, 0, \ldots, 0]$$

This all-zero feature vector provides no discriminative information to distinguish the intermittent faulty processor u_i from a truly fault-free processor u in a fault-free neighborhood. The PMC testing results would be identical in both cases, making accurate diagnosis difficult.

To address this limitation, we conduct R independent testing rounds. The intuition is that across multiple rounds, an intermittent faulty node is highly likely to manifest its fault, creating detectable patterns in the aggregated test results. In each testing round r, we need to know whether each node is currently exhibiting faulty behavior or normal behavior. For any node u, its current state $S_u^{(r)}$ is determined as follows:

$$S_u^{(r)} = \begin{cases} \text{faulty}, & \text{if } u = u_i \text{ and manifests fault} \\ \text{fault-free}, & \text{otherwise} \end{cases}$$

Specifically, for each intermittent faulty node, the fault manifests unpredictably in each round, while normal nodes always remain fault-free. This means that across multiple rounds of testing, an intermittent faulty node will sometimes appear fault-free and sometimes appear faulty, creating the sporadic behavior characteristic of intermittent faults.

For each testing round r, when node u tests its neighbor v, the test result $T_{u,v}^{(r)}$ depends on the current states of both nodes:

$$T_{u,v}^{(r)} = \begin{cases} 0, & \text{if } S_u^{(r)} \text{ and } S_v^{(r)} \text{ are faulty-free} \\ 1, & \text{if } S_u^{(r)} \text{ is faulty-free and } S_v^{(r)} \text{ is faulty} \\ \{0, 1\}, & \text{if } S_u^{(r)} \text{ is faulty} \end{cases}$$

This means that only fault-free nodes can provide reliable diagnostic information. When a node is currently exhibiting faulty behavior, its test results are random 0 or 1 values regardless of the tested node's actual state.

B. Locality-Aware Feature Extraction

After conducting R testing rounds, we obtain a collection of syndrome results $\{T_{u,v}^{(1)}, T_{u,v}^{(2)}, \ldots, T_{u,v}^{(R)}\}$ for each node pair (u, v). The next challenge is transforming these raw test results into meaningful features that can effectively distinguish intermittent faulty nodes from normal nodes.

Our approach leverages the PMC model's locality principle—each node can only observe its immediate neighbors—to construct node-level features based on local testing patterns. The key insight is that the proportion of positive test results across multiple rounds creates distinctive signatures for different node types.

1) Proportion-Based Statistical Features: After R testing rounds, we aggregate the syndrome matrices into a proportion matrix. For each node pair (u, v):

$$P_{u,v} = \frac{1}{R} \sum_{r=1}^{R} T_{u,v}^{(r)}$$

This proportion $P_{u,v}$ represents the frequency with which node u's test of neighbor v yields a positive result. These proportions create distinctive patterns that enable fault classification.

For each node u, we construct its feature vector by collecting test proportions to all its neighbors:

$$\mathbf{f}_u = [P_{u,v_1}, P_{u,v_2}, \ldots, P_{u,v_{|N(u)|}}]$$

where $N(u) = \{v_1, v_2, \ldots, v_{|N(u)|}\}$ represents the neighborhood of node u.

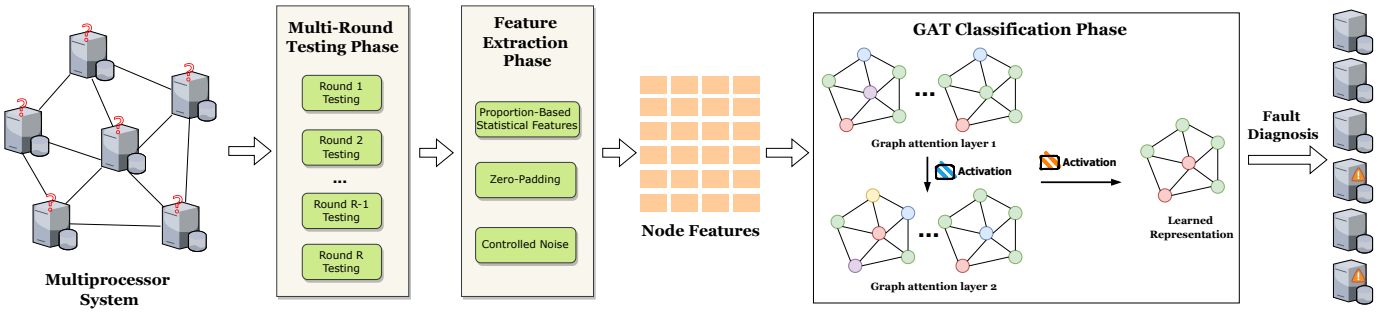

Fig. 2: Overview of FAULTGAT

This feature vector encodes the local diagnostic context around each node. A fault-free node will have features reflecting the true fault states of its neighbors, while a faulty node will have features dominated by random values, creating a clear distinction for classification. The locality constraint ensures that each node's diagnosis relies only on information it can directly observe, maintaining consistency with the PMC model.

2) Handling Variable Node Degrees: Within a network graph, different nodes may have different numbers of neighbors (i.e., different degrees). To ensure consistent feature dimensions across all nodes, we apply zero-padding to equalize feature vector lengths:

$$\tilde{\mathbf{f}}_u = [\mathbf{f}_u, \underbrace{0, 0, \ldots, 0}_{d_{\max} - |N(u)|}]$$

where $d_{\max} = \max_{v \in V} |N(v)|$ is the maximum degree in the network. This allows nodes with fewer neighbors to have the same feature dimension as nodes with more neighbors, enabling uniform processing by the neural network.

3) Feature Enhancement for Training Stability: Many fault-free nodes in fault-free neighborhoods produce all-zero feature vectors, which can cause training instability. We add very small controlled noise to enhance feature diversity while preserving the underlying diagnostic patterns:

$$\hat{f}_{u,v} = \text{clip}_{[0,1]}(\tilde{f}_{u,v} + \epsilon_{u,v})$$

where $\epsilon_{u,v} \sim N(0, \sigma^2)$ with $\sigma \ll P_{u,v}$ for any non-zero proportion $P_{u,v}$. The noise magnitude is deliberately kept much smaller than typical fault manifestation proportions to ensure that the neural network can still distinguish between noise and actual fault symptoms. This preserves the statistical properties while improving model trainability.

Figure 3 illustrates a simplified FAULTGAT node feature extraction process: $Node_1$ is intermittently faulty, with all four of its neighboring servers being healthy. When extracting the node feature for Server 1, R rounds of testing (assuming R=10) are conducted on all neighboring nodes, collecting test results summed as [4, 5, 5, 3]. These are then used to compute the probabilities of test result 1 when the node is in a faulty state, yielding [0.4, 0.5, 0.5, 0.3], which forms the node feature used by FAULTGAT. (Since no zero elements appear, the addition of controlled noise is not demonstrated in this example.)

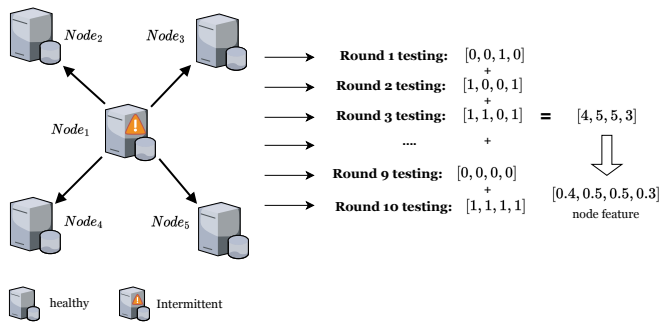

Fig. 3: An simplified example of the FAULTGAT node feature extraction process

C. Graph Attention Network Architecture

The extracted node features are processed through a GAT architecture designed to leverage both local neighborhood information and global network structure for fault classification.

1) Feature Transformation Layer: Raw proportion-based features are first enhanced through a transformation layer:

$$\mathbf{X}^{(0)} = \text{LeakyReLU}(\text{LayerNorm}(\mathbf{F}\mathbf{W}_{ft} + \mathbf{b}_{ft}))$$

where $\mathbf{F} \in \mathbb{R}^{n \times d_{\max}}$ contains all node features and $\mathbf{W}_{ft} \in \mathbb{R}^{d_{\max} \times d_h}$ projects them to hidden dimension d_h. The LayerNorm operation stabilizes training while LeakyReLU provides non-linearity.

2) Multi-Head Attention for Adaptive Information Fusion: The GAT layers use multi-head attention to adaptively weight information from different neighbors. For each attention head k in layer l:

$$e_{uv}^{(k)} = \text{LeakyReLU}(\mathbf{a}_k^T[\mathbf{W}_k^{(l)}\mathbf{h}_u^{(l-1)} \parallel \mathbf{W}_k^{(l)}\mathbf{h}_v^{(l-1)}])$$

$$\alpha_{ij}^{(k)} = \frac{\exp(e_{uv}^{(k)})}{\sum_{m \in \mathcal{N}(i)} \exp(e_{im}^{(k)})}$$

$$\mathbf{h}_u^{(k,l)} = \sum_{v \in \mathcal{N}(u)} \alpha_{uv}^{(k)} \mathbf{W}_k^{(l)} \mathbf{h}_v^{(l-1)}$$

The attention weights $\alpha_{uv}^{(k)}$ enable the model to focus on the most reliable diagnostic information. When diagnosing node u, the model learns to assign higher weights to neighbors that are likely fault-free and thus provide reliable test results, while de-emphasizing information from potentially faulty neighbors.

3) Multi-Layer Architecture: Our model employs L GAT layers with K attention heads each. For intermediate layers ($l = 1, \ldots, L-1$):

$$\mathbf{H}^{(l)} = \text{Concat}(\mathbf{h}^{(1,l)}, \mathbf{h}^{(2,l)}, \ldots, \mathbf{h}^{(K,l)})$$
$$\mathbf{H}^{(l)} = \text{Dropout}(\text{ReLU}(\text{BatchNorm}(\mathbf{H}^{(l)})))$$

The final layer produces binary classification logits:

$$\mathbf{Y} = \text{GAT}^{(L)}(\mathbf{H}^{(L-1)}, \mathcal{E})$$

where $\mathbf{Y} \in \mathbb{R}^{n \times 2}$ contains logits for each node's fault status.

Algorithm 1: GAT-based Intermittent Fault Diagnosis

Input: Multiprocess system $G = (V, E)$, test rounds R
Output: Node classifications $\hat{\mathbf{y}}$

Multi-Round Testing Phase
for $r = 1$ to R **do**
  for each edge $(u, v) \in E$ **do**
    $T_{u,v}^{(r)} \leftarrow \text{PMCTest}(u, v)$
  end for
end for
Feature Extraction Phase
for each node $u \in V$ **do**
  for each neighbor $v \in N(u)$ **do**
    $P_{u,v} \leftarrow \frac{1}{R} \sum_{r=1}^{R} T_{u,v}^{(r)}$
  end for
  $\mathbf{f}_u \leftarrow [P_{u,v} : v \in N(u)]$
  $\tilde{\mathbf{f}}_u \leftarrow \text{ZeroPad}(\mathbf{f}_u, d_{\max})$
  $\hat{\mathbf{f}}_u \leftarrow \text{AddNoise}(\tilde{\mathbf{f}}_u)$
end for
GAT Classification Phase
$\mathbf{X}^{(0)} \leftarrow \text{FeatureTransform}(\{\hat{\mathbf{f}}_u\})$
for $l = 1$ to $L-1$ **do**
  $\mathbf{H}^{(l)} \leftarrow \text{MultiHeadGAT}^{(l)}(\mathbf{H}^{(l-1)}, E)$
  $\mathbf{H}^{(l)} \leftarrow \text{Dropout}(\text{ReLU}(\text{BatchNorm}(\mathbf{H}^{(l)})))$
end for
$\mathbf{Y} \leftarrow \text{GAT}^{(L)}(\mathbf{H}^{(L-1)}, E)$
$\hat{\mathbf{y}} \leftarrow \arg\max(\text{Softmax}(\mathbf{Y}))$
return $\hat{\mathbf{y}}$

D. Training Strategy and Optimization

1) Loss Function and Class Balancing: The model is trained using cross-entropy loss to handle binary classification:

$$\mathcal{L} = -\frac{1}{n} \sum_{u=1}^{n} [y_u \log \sigma(\mathbf{Y}_u)_1 + (1 - y_i) \log \sigma(\mathbf{Y}_u)_0]$$

where $y_u \in \{0, 1\}$ is the ground truth label, σ is the softmax function, and $\mathbf{Y}_u$ contains the logits for node u.

2) Optimization and Regularization: We employ the Adam optimizer with learning rate scheduling and weight decay. Key regularization techniques include:

- Gradient clipping: Prevents exploding gradients during training,
- Early stopping: Training terminates when validation F1-score plateaus,
- Dropout: Applied after each GAT layer for regularization,
- Batch normalization: Stabilizes training and improves convergence.

Algorithm 1 describes our complete GAT-based intermittent fault diagnosis framework, which consists of three main phases. The *Multi-Round Testing Phase* conducts R independent testing rounds following the PMC model to collect syndrome data across multiple observations. The *Feature Extraction Phase* transforms the raw test results into proportion-based node features, applying zero-padding for uniform dimensions and noise augmentation for training stability. Finally, the *GAT Classification Phase* employs multi-layer Graph Attention Networks to learn from the local neighborhood patterns and classify each node as either fault-free or intermittent faulty.

IV. Simulation

In this section, we aim to answer the following research questions (RQs) to thoroughly evaluate the performance and effectiveness of FaultGAT:

- **RQ1**: Can FaultGAT accurately diagnose faults within theoretical diagnosability bounds?
- **RQ2**: How does FaultGAT perform when the number of faults exceeds diagnosability bounds?
- **RQ3**: How does FaultGAT perform under partial syndromes?

A. Evaluated Topologies

The FaultGAT framework is designed to operate effectively across general graph topologies, without imposing specific structural requirements on the underlying network. In this paper, we evaluate its performance on two representative interconnection networks: a simple graph, the Hypercube Q_n, and a complex graph, the Augmented k-ary n-cube $AQ_{n,k}$. These topologies allow us to demonstrate FaultGAT's versatility in handling both straightforward and intricate network structures under the PMC model for intermittent fault diagnosis.

B. FaultGAT *model setting*

The FaultGAT model is configured with 2 graph attention layers, each using 8 attention heads and a hidden dimension of 64. The input features are first processed through a feature transformation layer with layer normalization and LeakyReLU activation. Batch normalization is applied after each attention layer, and dropout with probability 0.3 is used during training for regularization. The FaultGAT is trained using the Adam optimizer with an initial learning rate of 0.002 and FocalLoss (alpha=0.25, gamma=2.0). Training runs for a maximum of 200-300 epochs with a batch size of 16, incorporating early stopping based on validation performance to prevent overfitting.

C. Baseline

To evaluate the performance of our proposed diagnostic tool, FaultGAT, we compare it with the RNNIFDCom algorithm from [14]. RNNIFDCom is a state-of-the-art method for intermittent fault diagnosis in networks under the comparison

model (MM model), making it a relevant baseline for assessing system-level fault identification. However, as [14] does not provide source code, we implemented RNNIFDCOM based on the descriptions in the paper. To ensure a fair comparison, we adapted the implementation to operate under PMC model instead of the comparison model, aligning it with the diagnostic framework used by FAULTGAT.

RNNIFDCOM employs a multilayer feedforward neural network with backpropagation to identify faulty nodes using partial syndromes, addressing the challenge of incomplete test data in large-scale networks. The basic principle involves training the network to map test outcomes to node fault statuses. The input layer, of size $|C(G)|$, represents the test outcomes from the diagnostic multigraph $D_G = (V(G), C(G))$, where $|C(G)|$ is the number of test edges, test outcomes are encoded as -0.5 (fault-free) or 0.5 (faulty). The output layer, of size $|V(G)|$, corresponds to the number of nodes in the network G, with each output x_i^L indicating the fault status of node v_i (close to 1 for faulty, -1 for fault-free).

The configuration of RNNIFDCOM is as follows: A multilayer feedforward neural network with L layers: one input layer, $L-1$ hidden layers, and one output layer. The activation function is $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, mapping inputs to $(-1, 1)$. Due to the absence of specific details on the number of hidden layers and neurons in [14], we adopted the configuration from [13], which shares a similar neural network architecture. For a network with $n = |V(G)|$ nodes, the number of hidden layers is set as $L - 1 = 3$ if $n \leq 50$, $L - 1 = 4$ if $50 < n \leq 100$, and $L - 1 = 5$ if $n > 100$. Each hidden layer contains $\max(n, \lfloor n \cdot \log(\max(2, n \mod 10))\rfloor)$ neurons, or $\max(n, \lfloor n \cdot \log(\max(2, n))/10\rfloor)$ if $n \mod 10 = 0$.

D. DataSet

The dataset was generated following a methodology similar to that proposed in [14]. Specifically, we generated different t-diagnosable and non-diagnosable hypercube and the augmented k-ary n-cube interconnection topologies.

Since FAULTGAT processes node features derived from R rounds of testing, obtaining a complete node feature requires testing all neighboring nodes R times, with the final feature representing an aggregation of these R test rounds. In contrast, RNNIFDCOM takes as input the results of a single test round across all nodes. Therefore, for a fair comparison, the RNNIFDCOM dataset should be R times larger than that of FAULTGAT. In the experiments, we set the number of test rounds to $R = 10$, which means that if FAULTGAT uses 200 graph instances for training, RNNIFDCOM should uses 2,000 samples (200 graphs × 10 test rounds) to process equivalent information.

We conducted three research questions with the following experimental configurations:

- For RQ1, we generated 200 graphs per dimension for both Q_n and $AQ_{n,k}$ topologies with 10 repeated runs, training each model for 200 epochs. The intermittent faulty nodes were randomly generated within the theoretical intermittent fault diagnosability limit t_i.
- For RQ2, we generated 300 graphs per dimension for both Q_n and $AQ_{n,k}$ topologies with 10 repeated runs, training each model for 300 epochs. The intermittent faulty nodes were generated beyond the theoretical diagnosability limit using proportional fault injection. We tested fault ratios from 10% to 50% with 1% increments.
- For RQ3, we generated 300 graphs per dimension for both Q_n and $AQ_{n,k}$ topologies with 10 repeated runs, training each model for 300 epochs. We operated within the theoretical diagnosability limit while progressively disabling nodes. Starting with 5% disabled nodes, we increased the ratio by 5% increments up to 50%. Disabled nodes could neither serve as testers nor testees, resulting in partial syndrome for any node affected by disabled nodes.

For all experiments, the dataset was split into training (70%), validation (15%), and testing (15%) sets for each experimental run.

E. Metrics

We evaluate the performance of both FAULTGAT and RNNIFDCOM using four key metrics: *Accuracy*, *Precision*, *Recall*, and *F1-score*. ***All metrics are calculated with respect to faulty nodes as the positive class and fault-free nodes as the negative class.*** Accuracy measures the overall correctness of the fault diagnosis model, representing the proportion of correctly classified nodes (both faulty and fault-free) among all nodes:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures the reliability of positive predictions, indicating the proportion of nodes predicted as faulty that are actually faulty:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures the completeness of positive predictions, indicating the proportion of actual faulty nodes that are correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-score provides a harmonic mean of precision and recall, offering a balanced measure that considers both false positives and false negatives:

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times TP}{2 \times TP + FP + FN}$$

Where the symbols represent:

- TP (True Positive): Faulty nodes correctly identified as faulty
- TN (True Negative): Fault-free nodes correctly identified as fault-free
- FP (False Positive): Fault-free nodes incorrectly identified as faulty
- FN (False Negative): Faulty nodes incorrectly identified as fault-free

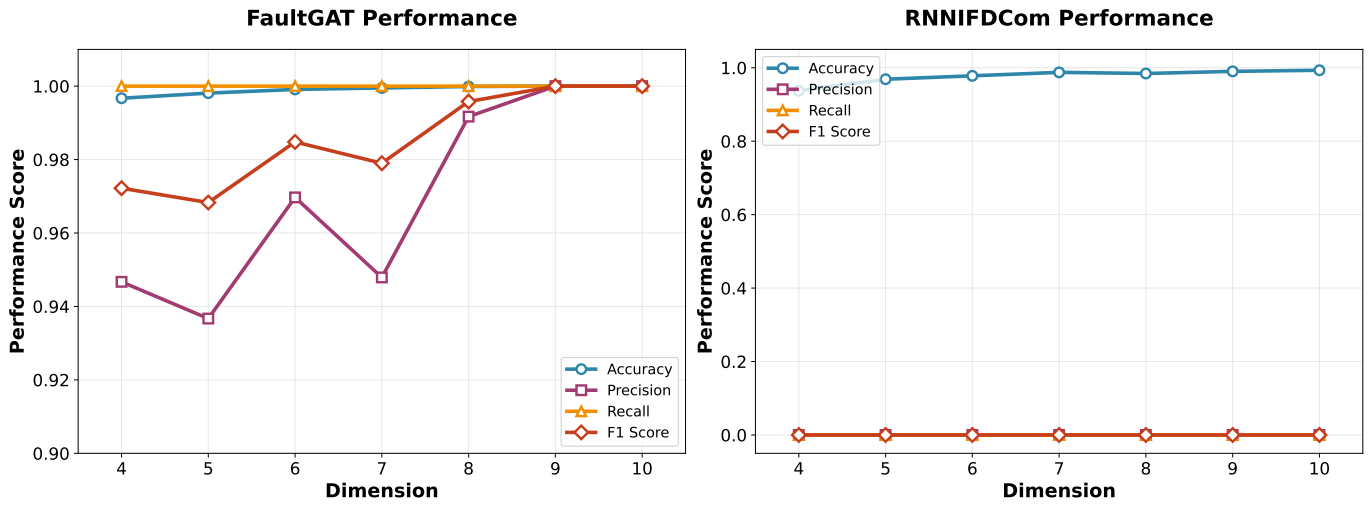

Fig. 4: Performance of FAULTGAT and RNNIFDCOM for Q_n within Theoretical Diagnosability Bounds

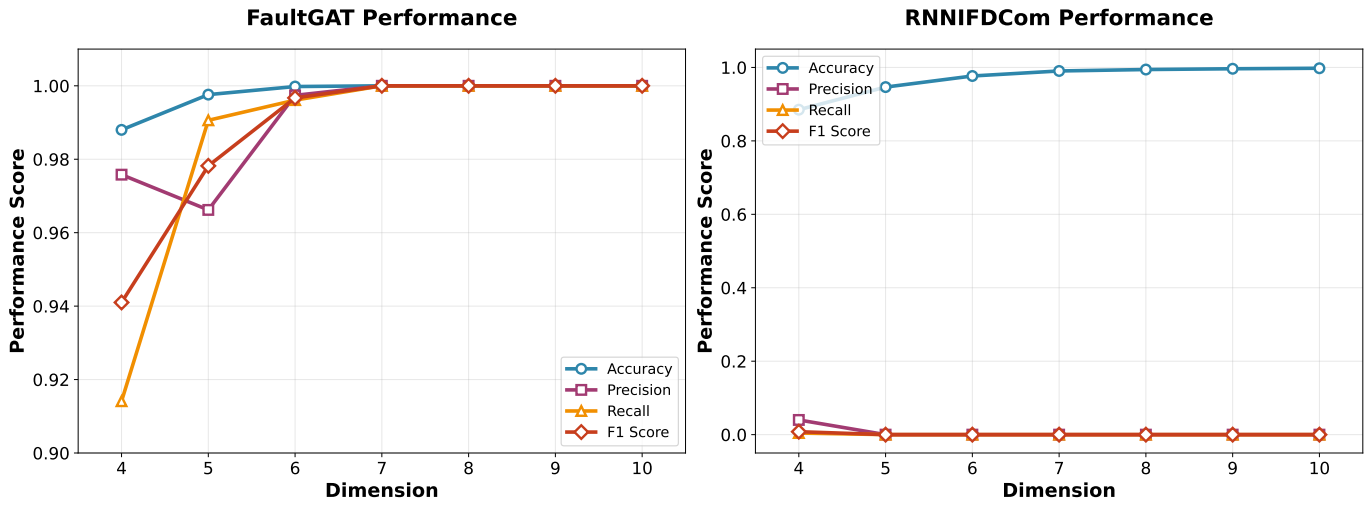

Fig. 5: Performance of FAULTGAT and RNNIFDCOM for $AQ_{n,k}$ within Theoretical Diagnosability Bounds

F. Evaluation environment

All experiments were conducted on a computing platform equipped with an Intel Xeon processor, dual NVIDIA GeForce RTX 4090 GPUs with CUDA support, and 32GB RAM. The software environment included Ubuntu 20.04, Python 3.10, PyTorch Geometric 2.6.0.

G. RQ1: Theoretical Diagnosability Performance Analysis

Figures 4 and 5 present the results of the FAULTGAT and RNNIFDCOM models for intermittent fault diagnosis within theoretical diagnosability bounds on the hypercube Q_n and the augmented k-ary n-cube $AQ_{n,k}$. These figures illustrate the performance of both models across four key metrics—accuracy, precision, recall, and F1-score—for topology dimensions ranging from $n = 4$ to $n = 10$. The experiments assess the models' ability to diagnose intermittent faults under the PMC model, with fault counts constrained within the theoretical diagnosability limits ($t_i(Q_n) = n - 2$ and $t_i(AQ_{n,k}) = 4n - 6$).

For hypercubes Q_n in Fig. 4, FAULTGAT maintains an accuracy above 99.67% across all tested dimensions, with performance reaches 100% at $n = 9, 10$. The recall metric consistently reaches 1.0, indicating complete detection of faulty nodes. Precision ranges from 94.67% to 100%, and the F1-score varies between 97.22% and 100%, demonstrating balanced diagnostic performance. For the augmented k-ary n-cubes $AQ_{n,k}$ in Fig. 5, FAULTGAT exhibits stronger performance. Accuracy starts at 98.80% for smaller networks and reaches 100% for dimensions $n = 7, 8, 9, 10$. Precision scales from 97.58% to 100%, and recall increases from 91.41% to 100%, resulting in F1-scores that also achieve 1.0 in larger configurations. The enhanced connectivity of $AQ_{n,k}$ provides additional structural information, which FAULTGAT 's graph attention mechanism leverages effectively, leading to improved performance in higher-dimensional topologies.

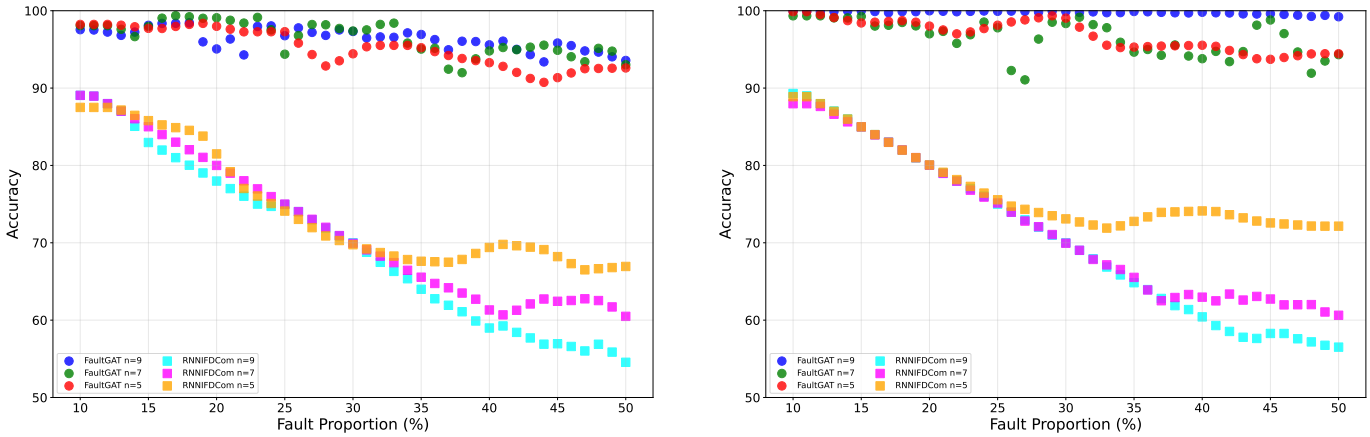

Fig. 6: Accuracy of Q_n (left) and $AQ_{n,k}$ (right) beyond Theoretical Diagnosability Bounds

RNNIFDCOM shows a different performance profile. For Q_n, its accuracy ranges from 93.75% to 99.32%, and for $AQ_{n,k}$, it ranges from 88.49% to 99.78%. These accuracy figures appear competitive but are misleading, as precision, recall, and F1-scores are near zero across both topologies, with only minimal non-zero values (e.g., 0.40% precision and 0.44% recall) in the smallest $AQ_{n,k}$ configuration. This indicates that RNNIFDCOM struggles to identify intermittent faulty nodes, predominantly classifying nodes as fault-free.

The high accuracy of RNNIFDCOM is primarily due to the class imbalance inherent in the theoretical diagnosability bounds, where the number of faulty nodes is limited ($t_i(Q_n) = n-2$ and $t_i(AQ_{n,k}) = 4n-6$), resulting in a small proportion of intermittent faulty nodes relative to the total node count. RNNIFDCOM converges to a trivial solution, predicting nearly all nodes as fault-free, which yields high accuracy because fault-free nodes dominate but leads to near-zero precision, recall, and F1-scores. This behavior suggests that RNNIFDCOM fails to distinguish between intermittent faulty and fault-free nodes, as its sequence-based architecture does not effectively capture the localized fault patterns inherent in graph-structured networks.

Answer to RQ1: FAULTGAT effectively diagnoses intermittent faults within the theoretical diagnosability bounds in hypercubes and augmented k-ary n-cubes, demonstrating strong performance across all metrics and detecting nearly all faulty nodes, with improved performance in larger, more complex networks.

H. RQ2: Fault Diagnosis beyond Theoretical Diagnosability Bounds

Figure 6 and Tables I and II present the performance evaluation of FAULTGAT and RNNIFDCOM for intermittent fault diagnosis when the number of faulty nodes exceeds the theoretical diagnosability limits. The experiments test fault ratios from 10% to 50% across network dimensions $n = 5, 7, 9$ for both Q_n and $AQ_{n,k}$ topologies. Figure 6 illustrates accuracy trends, while Tables I and II detail precision, recall, and F1-score metrics.

TABLE I: Performance Comparison of FAULTGAT and RNNIFDCOM on Q_n Across Different Network Sizes and Fault Ratios (Partial)

Dimension	Fault Ratio	FAULTGAT			RNNIFDCOM		
		Precision	Recall	F1-score	Precision	Recall	F1-score
$n = 5$	0.10	0.9881	0.9222	0.9540	0.0000	0.0000	0.0000
	0.20	0.9609	0.9556	0.9582	0.0000	0.0000	0.0000
	0.30	0.9813	0.8733	0.9242	0.5423	0.0727	0.1282
	0.40	0.9400	0.9215	0.9307	0.6402	0.5895	0.6138
	0.50	0.8807	0.9229	0.9268	0.6707	0.6654	0.6681
$n = 7$	0.10	0.9900	0.9857	0.9878	0.0000	0.0000	0.0000
	0.20	0.9794	0.9769	0.9782	0.0000	0.0000	0.0000
	0.30	0.9710	0.9386	0.9545	0.4561	0.0237	0.0450
	0.40	0.9505	0.9346	0.9425	0.5214	0.3191	0.3959
	0.50	0.9605	0.9104	0.9348	0.5456	0.5449	0.5452
$n = 9$	0.10	0.9900	0.9857	0.9878	0.0000	0.0000	0.0000
	0.20	0.9890	0.9256	0.9563	0.0000	0.0000	0.0000
	0.30	0.9921	0.8632	0.9232	0.4049	0.0113	0.0220
	0.40	0.9941	0.8850	0.9364	0.4645	0.2299	0.3076
	0.50	0.9605	0.9104	0.9348	0.5456	0.5449	0.5452

TABLE II: Performance Comparison of FAULTGAT and RNNIFDCOM on $AQ_{n,k}$ Across Different Network Sizes and Fault Ratios (Partial)

Dimension	Fault Ratio	FAULTGAT			RNNIFDCOM		
		Precision	Recall	F1-score	Precision	Recall	F1-score
$n = 5$	0.10	0.9890	1.0000	0.9945	0.0000	0.0000	0.0000
	0.20	0.9795	0.9933	0.9864	0.0000	0.0000	0.0000
	0.30	0.9876	0.9958	0.9917	0.6773	0.1854	0.2911
	0.40	0.9849	0.9933	0.9891	0.7334	0.6903	0.7112
	0.50	0.9876	0.9458	0.9662	0.7130	0.7056	0.7093
$n = 7$	0.10	0.9974	0.9930	0.9952	0.1333	0.0148	0.0267
	0.20	0.9809	0.9975	0.9891	0.0000	0.0000	0.0000
	0.30	0.9941	0.9361	0.9642	0.4859	0.0431	0.0791
	0.40	0.9698	0.9591	0.9644	0.5433	0.4576	0.4968
	0.50	0.9784	0.9550	0.9633	0.6022	0.5972	0.5997
$n = 9$	0.10	1.0000	0.9972	0.9986	0.0000	0.0000	0.0000
	0.20	1.0000	0.9967	0.9984	0.0000	0.0000	0.0000
	0.30	0.9995	0.9986	0.9991	0.9000	0.0004	0.0008
	0.40	0.9973	0.9963	0.9968	0.5090	0.1893	0.2760
	0.50	0.9852	0.9810	0.9831	0.5634	0.5622	0.5628

As shown in the left subgraph of Figure 6, FAULTGAT maintains accuracy above 90% across all fault ratios and dimensions for Q_n. Tables I provides detailed metrics: for $n = 5$, FAULTGAT achieves precision of 0.9881, recall of 0.9222, and F1-score of 0.9540 at a 10% fault ratio. Even when the fault ratio increases to 50%, the performance shows only a slight decrease with precision of 0.8807, recall of 0.9229,

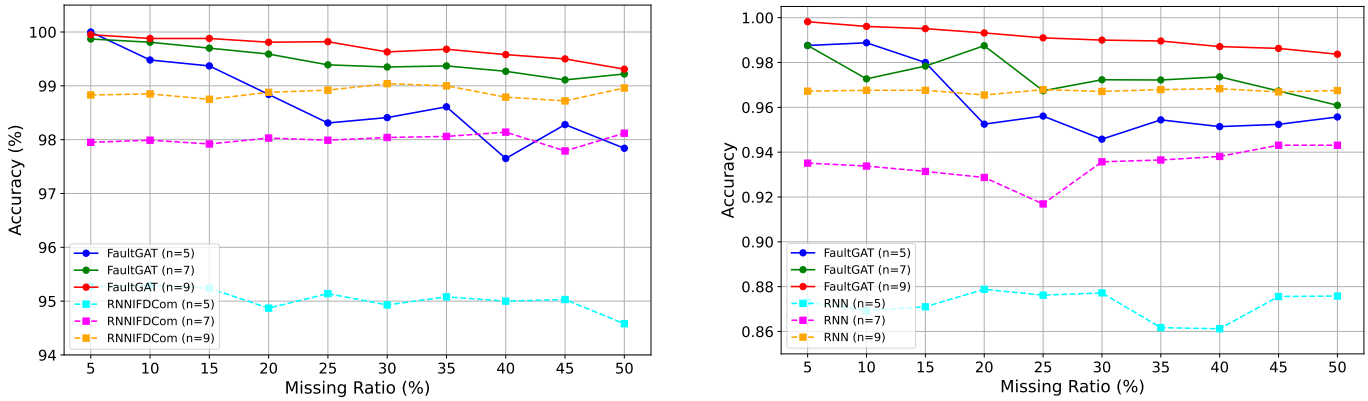

Fig. 7: Accuracy of Q_n (left) and $AQ_{n,k}$ (right) with partial syndrome

and F1-score of 0.9268. For $n = 7$ and $n = 9$, FAULTGAT 's metrics remain strong, with precision ranging from 0.9505 to 0.9900, recall from 0.8632 to 0.9857, and F1-scores from 0.9232 to 0.9878. In contrast, Tables I reveals RNNIFDCOM's severe limitations in fault detection. At low fault ratios (0.10 and 0.20), RNNIFDCOM exhibits precision, recall, and F1-scores approaching 0.0 across all dimensions for Q_n. While these metrics improve slightly as the fault ratio increases, this apparent improvement is misleading when examined alongside Figure 6, which shows RNNIFDCOM's accuracy declining sharply and reaching approximately 50% at a 50% fault ratio. This contradictory pattern reveals that RNNIFDCOM essentially fails to learn the distinction between intermittent faulty nodes and fault-free nodes, instead tending to classify all nodes as fault-free.

For $AQ_{n,k}$, the right subgraph of Figure 6 indicates FAULTGAT 's accuracy remains above 92% across all tested fault ratios and dimensions. Table II shows FAULTGAT achieving precision, recall, and F1-scores above 0.97 for $n = 5$ at lower fault ratios (e.g., 10%), with minor declines to 0.94–0.96 at 50%. For $n = 7$ and $n = 9$, FAULTGAT sustains precision above 0.97, recall above 0.93, and F1-scores above 0.96, capitalizing on $AQ_{n,k}$'s enhanced connectivity to improve fault detection. For RNNIFDCOM, it exhibits similar poor performance on $AQ_{n,k}$ as observed with Q_n. Tables II shows that precision, recall, and F1-scores of $AQ_{n,k}$ approach 0.0 at low fault ratios and only marginal improvements at higher fault ratios. Consistent with the Q_n results, Figure 6 confirms that RNNIFDCOM's accuracy drops to around 50% at a 50% fault ratio for $AQ_{n,k}$ as well. This consistent degradation across both Q_n and $AQ_{n,k}$ demonstrates RNNIFDCOM's fundamental inability to do nodes classification tasks effectively.

Answer to RQ2: FAULTGAT effectively diagnoses intermittent faults beyond theoretical diagnosability bounds in hypercube and augmented k-ary n-cube topologies, achieving strong performance even when the proportion of faulty nodes reaches 50%, with accuracy exceeding 90%.

TABLE III: Performance Comparison of FAULTGAT and RNNIFDCOM on Q_n Across Different Network Sizes and Missing Ratios (Partial)

Dimension	Missing Ratio	FAULTGAT			RNNIFDCOM		
		Precision	Recall	F1-score	Precision	Recall	F1-score
$n = 5$	0.10	0.9857	0.9857	0.9986	0.0000	0.0000	0.0000
	0.20	1.0000	0.7609	0.8642	0.0000	0.0000	0.0000
	0.30	0.9787	0.6765	0.8000	0.0000	0.0000	0.0000
	0.40	1.0000	0.5111	0.6765	0.0000	0.0000	0.0000
	0.50	0.9999	0.5556	0.7143	0.0000	0.0000	0.0000
$n = 7$	0.10	1.0000	0.9054	0.9504	0.0000	0.0000	0.0000
	0.20	0.9999	0.7973	0.8872	0.0000	0.0000	0.0000
	0.30	0.9998	0.6761	0.8067	0.0000	0.0000	0.0000
	0.40	1.0000	0.6377	0.7788	0.0000	0.0000	0.0000
	0.50	0.9787	0.5476	0.7023	0.0000	0.0000	0.0000
$n = 9$	0.10	1.0000	0.8977	0.9461	0.0000	0.0000	0.0000
	0.20	1.0000	0.8353	0.9103	0.0000	0.0000	0.0000
	0.30	1.0000	0.6790	0.8088	0.0000	0.0000	0.0000
	0.40	0.9999	0.6375	0.7786	0.0000	0.0000	0.0000
	0.50	1.0000	0.4177	0.5893	0.0000	0.0000	0.0000

TABLE IV: Performance Comparison of FAULTGAT and RNNIFDCOM on $AQ_{n,k}$ Across Different Network Sizes and Missing Ratios (Partial)

Dimension	Missing Ratio	FAULTGAT			RNNIFDCOM		
		Precision	Recall	F1-score	Precision	Recall	F1-score
$n = 5$	0.10	0.9688	0.9394	0.9538	0.0000	0.0000	0.0000
	0.20	0.8020	0.8182	0.8100	0.0000	0.0000	0.0000
	0.30	0.8800	0.6600	0.7543	0.0000	0.0000	0.0000
	0.40	0.6630	0.6100	0.6354	0.7500	0.0042	0.0084
	0.50	0.6104	0.4747	0.5341	1.0000	0.0018	0.0036
$n = 7$	0.10	1.0000	0.8113	0.8958	0.0000	0.0000	0.0000
	0.20	0.7640	0.8500	0.8047	0.0000	0.0000	0.0000
	0.30	0.8456	0.7188	0.7770	0.0000	0.0000	0.0000
	0.50	0.9231	0.4557	0.6102	0.0000	0.0000	0.0000
	0.40	0.5062	0.5157	0.5109	0.0000	0.0000	0.0000
$n = 9$	0.10	0.9679	0.9095	0.9378	0.0000	0.0000	0.0000
	0.20	0.9946	0.7991	0.8862	0.0000	0.0000	0.0000
	0.30	1.0000	0.6996	0.8232	0.0000	0.0000	0.0000
	0.40	0.9924	0.6103	0.7558	0.0000	0.0000	0.0000
	0.50	0.9744	0.5327	0.6888	0.0000	0.0000	0.0000

I. RQ3: Fault Diagnosis with partial syndrome

Figure 7 and Tables III and IV present the performance evaluation of FAULTGAT and RNNIFDCOM for intermittent fault diagnosis under partial syndrome conditions in hypercube and augmented k-ary n-cube topologies with dimensions $n = 5, 7, 9$. The experiments test missing ratios from 5% to 50%, assessing the models' robustness when syndrome is incomplete. Figure 7 illustrates accuracy trends, while

Tables III and IV detail precision, recall, and F1-score metrics.

For Q_n, the left subgraph of Figure 7 demonstrates FAULTGAT's better performance with consistently high accuracy above 99% for $n = 7$ and $n = 9$, and above 97% for $n = 5$ across all missing ratios. In contrast, RNNIFDCOM's accuracy performance achieves above 98% for $n = 9$, above 97% for $n = 7$, and above 94% for $n = 5$. Table III further reveals that FAULTGAT achieves better performance at low missing ratios, with F1-scores ranging from 0.9461 to 0.9986 at 10% missing ratio across all dimensions, and maintains strong performance even as missing ratios increase to 50% where F1-scores range from 0.5893 to 0.7143. Notably, FAULTGAT demonstrates precision consistently near or at 1.0000 across all dimensions and missing ratios, while recall shows more substantial decline from approximately 0.90 at 10% missing ratio to 0.42-0.56 at 50% missing ratio, with larger dimensions generally maintaining better recall performance. However, RNNIFDCOM shows consistently zero precision, recall, and F1-scores across all tested conditions in Table III, indicating significant limitations in handling partial syndromes in hypercube topologies compared to FAULTGAT.

For $AQ_{n,k}$, the right subgraph of Figure 7 demonstrates FAULTGAT's superior performance with consistently high accuracy above 94% across all dimensions and missing ratios, with $n = 9$ achieving accuracy above 98%. In contrast, RNNIFDCOM's accuracy performance varies considerably with network dimension, achieving above 96% for $n = 9$, above 92% for $n = 7$, and above 86% for $n = 5$. Table IV further reveals that FAULTGAT achieves good performance at low missing ratios, with F1-scores ranging from 0.8958 to 0.9538 at 10% missing ratio across all dimensions, but experiences consistent decline as missing ratios increase to 50% where F1-scores drop to between 0.5341 and 0.6888. Notably, larger network dimensions exhibit better fault tolerance, with $n = 9$ maintaining high precision above 0.96 even at severe missing ratios while $n = 5$ shows more dramatic precision degradation from 0.9688 to 0.6104 and both precision and recall decline with increasing missing ratios. However, RNNIFDCOM demonstrates significantly inferior performance with consistently low precision, recall, and F1-scores across all tested conditions, highlighting its inability to handle partial syndromes in complex topologies compared to FAULTGAT's consistently robust performance regardless of network size or missing data severity.

The performance differences between topologies can be attributed to their structural characteristics. For FAULTGAT, the relatively lower performance on $AQ_{n,k}$ compared to Q_n stems from the higher node connectivity in $AQ_{n,k}$, where each node has more neighbors than in hypercube topologies, making disabled nodes have a greater impact on syndrome completeness and consequently affecting fault diagnosis accuracy. For RNNIFDCOM, the near-zero precision, recall, and F1-scores in both topologies result from its sequence-based architecture's inherent limitations in capturing faulty node features, which become even more pronounced under partial syndrome conditions where the reduced symptom information further weakens its limited node classification capabilities.

Answer to RQ3: FAULTGAT effectively diagnoses intermittent faults under partial syndrome conditions in both hypercube and augmented k-ary n-cube topologies, achieving over 90% accuracy even with up to 50% of nodes disabled.

V. RELATED WORK

Intermittent fault diagnosis, a critical aspect of uncertain fault diagnosis, enhances network self-diagnosis capabilities. The inherent randomness and unpredictability of intermittent faults pose significant challenges to accurate diagnosis. Early research established foundational concepts and conditions for intermittent fault diagnosability [18]. Studies on specific network topologies, such as hypercubes [24], Crisp-cycle networks [25], product networks [26], and split-star networks [27], have developed necessary and sufficient conditions under models like PMC and MM, addressing scenarios with missing edges or simultaneous node and link failures [28]. However, these works mainly focus on theoretical proofs of intermittent fault diagnosability, often lacking practical algorithms for effective fault diagnosis. In this paper, we propose a practical method to diagnose intermittent fault nodes, extending beyond the constraints of theoretical diagnosability and applicable to general network topologies.

Several works have explored neural network-based approaches for fault diagnosis. Elhadef et al. [12], [13] proposed using artificial neural networks for diagnosing permanent fault diagnosis under the comparison model. More recently, Lin et al. [14] developed a similar neural network method for intermittent fault diagnosis under the comparison model. However, these neural network-based approaches typically use the entire graph's syndrome as input, requiring extensive training data, often involving tens of thousands of graph syndromes, to train their models. In contrast, our work employs a graph attention network that leverages the diagnostic symptom of each node for training, enabling more targeted learning and achieving better fault diagnosis performance with less experimental data.

VI. CONCLUSION

In this paper, we introduced FAULTGAT, a novel Graph Attention Network-based framework for diagnosing intermittent faults in interconnection networks under the PMC model. Our approach addresses critical limitations of traditional diagnostic algorithms by leveraging the locality-aware structure of GATs to perform node-level diagnosis using only local syndrome information. FAULTGAT transforms the challenging problem of intermittent fault diagnosis through a multi-round testing strategy and statistical feature extraction that effectively captures the sporadic nature of intermittent faults. Our extensive experimental evaluation across hypercube and augmented k-ary n-cube topologies demonstrates FAULTGAT's superior performance compared to existing approaches. Within theoretical diagnosability bounds, FAULTGAT achieves near 100% accuracy, while maintaining robust performance (>90% accuracy) even when fault counts exceed theoretical limits or when operating with partial syndromes. These results highlight FAULTGAT's potential as a practical and efficient solution for real-world fault diagnosis challenges in large-scale interconnection networks.

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